35 An α-particle and a β-particle have the same kinetic energy.

Why does the α-particle have a larger ionising effect than the β-particle as it passes through air?

A The α-particle has a larger charge and a larger velocity so is closer to an air particle for a shorter time.

B The α-particle has a larger charge and a smaller velocity so is closer to an air particle for a longer time.

C The α-particle has a smaller charge and a larger velocity so is closer to an air particle for a shorter time.

D The α-particle has a smaller charge and a smaller velocity so is closer to an air particle for a longer time.